Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/12/2016
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Cognition on Monozygotic Twin on Earth		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	12/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
Contact Email:	jbcharle@gmail.com		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2017 per PI (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinges, David Ph.D. (University of Pennsylvania) Gur, Ruben Ph.D. (University of Pennsylvania) Mollicone, Daniel Ph.D. (Pulsar Informatics, Inc.)		
Grant/Contract No.:	NNX14AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study is part of an integrated compilation of multiple studies led by multiple PIs at multiple centers which take advantage of a very unique opportunity to look at the effects of space travel on identical twins, one of whom experiences space travel for one year; the other remains Earth-bound for that same year. This study focuses on psychosocial and neurobehavioral experiments to investigate and characterize any differences between the flight subject and the ground subject in cognition, decision making ability, alertness levels, stress, and overall emotional well-being, as a result of the spaceflight environment. We will assess cognitive performance in the twins with the Cognition test battery 3 times pre-, 11 times in-, and 3 times post-flight. Cognition was specifically developed for high-performing astronauts and consists of 10 brief, validated neuropsychological tests that cover a wide range of cognitive domains. The project's specific aims serve to determine how cognitive aspects of the flight subject's phenotype change in response to acute and prolonged exposure to microgravity and other spaceflight stressors, both on a within-subject level (e.g., in-flight vs. pre-flight) and on a between-subjects level (i.e., in comparison to the ground subject's data). These phenotypic data will be shared within and outside the consortium and correlated with findings from the -omics analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Cognition test battery was specifically designed for the high performing and highly motivated astronaut population. It will thus be a useful tool for the objective assessment of cognitive performance in similar high performing populations on Earth (e.g., medical doctors).
Task Progress:	So far, we have administered all 10 tests of the Cognition battery 4 times pre-flight and 10 times in-flight in the twin that is performing the 12-month International Space Station (ISS) mission. We administered the same number of tests in the twin on Earth in close temporal proximity to the tests performed by the twin in space.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Basner M, Savitt A, Moore TM, Port AM, McGuire S, Ecker AJ, Nasrini J, Mollicone J, Mott CM, McCann T, Dinges DF, Gur RC. "Development and validation of the Cognition test battery for spaceflight." Aerospace Medicine and Human Performance. 2015 Nov;86(11):942-52. http://dx.doi.org/10.3357/AMHP.4343.2015 ; PubMed PMID: 26564759; PubMed Central PMCID: PMC4691281 , Nov-2015